

SMART INDIA HACKATHON 2026

SIH26063

Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal

COMPLETE TEAM HANDBOOK

Working Product Name

PolarSetu — Unified Discovery, Evidence and Outreach Layer for India's Polar Science

Purpose: give the whole team one source of truth for the problem, research, product scope, architecture, workflows, technology, data, AI behavior, 72-hour execution plan, six-slide SIH submission, demo story, judge Q&A, risks, and future roadmap.

Important: this is an engineering and presentation plan, not an official NCPOR product specification. Existing NCPOR systems are treated as current sources and integration targets, not as nonexistent infrastructure.

Prepared from the uploaded SIH 2026 idea-submission template, current public NCPOR resources, current public SIH26063 listings, current public GitHub examples, and current vendor/documentation pages. Research date: 27 September 2026.

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Use this contents page as the team navigation map; exact Word page numbers can change with edits.

0. Read This First — Team Decision in One Page

Problem in one sentence: India's polar programme already has valuable reports, datasets, publications, media and outreach material, but these capabilities are exposed through multiple official interfaces; SIH26063 asks for a comprehensive outreach portal that archives these resource types and generates website/social content.

Our strategy

Do not pitch “we made another repository.” Pitch “we built the discovery + evidence + outreach layer that connects polar resources, links related evidence, helps people understand it, and turns approved scientific information into audience-specific outreach.”

The core product loop is: Research → Knowledge → Outreach → Human Review → Public Impact. The repository is the foundation; the differentiator is how the resources are connected, explained and disseminated.

- MVP must work without any advanced ML training or complicated infrastructure.
- Backend is Go + Gin. Frontend is React + Vite + Tailwind. Database is PostgreSQL. Storage is Supabase Storage. LLM is Groq called from the Go server.
- Search starts with PostgreSQL Full-Text Search and pg_trgm. pgvector is optional only after the end-to-end demo works.
- Build a real domain experience: repository + expedition timeline/map + evidence-linked AI assistant + outreach studio + review queue.
- Use official NCPOR resources as references and links. Do not copy copyrighted images/files into the prototype unless permission/licence allows it.
- Never put API keys in the React bundle. AI calls happen in Go on the server.
- Never allow AI output to auto-publish in the demo. Human review is a visible product step.

Non-negotiable scope

Build	Do not build in the 72-hour window
Resource catalog	Elasticsearch / Kafka / microservices
Search	Complex ML training
Evidence-linked AI	Neo4j knowledge graph
Outreach drafts	Real social-media posting integrations
Expedition map/timeline	Real-time satellite processing
Admin review	Complex multi-tenant enterprise IAM
Deployment	Kubernetes

Deliverables at the end

- Working web prototype with seeded real-world-inspired data.
- Go REST API connected to PostgreSQL.
- Search, repository, expedition detail, map/timeline, AI assistant, outreach studio and approval workflow.
- A 6-slide SIH submission PDF using the supplied template.
- A 4–5 minute demo flow and a judge Q&A sheet.
- GitHub repository with README, architecture diagram, setup instructions and clean environment handling.

1. Problem Statement — Exact Meaning

Problem Statement ID: SIH26063

Title: Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal

Organization: Ministry of Earth Sciences (MoES)

Department: National Centre for Polar and Ocean Research (NCPOR)

Category: Software

A current public SIH2026 community archive shows 30 September 2026; verify the actual official portal/institutional submission window before relying on that date.

Official problem text

“Develop a comprehensive outreach portal that archives expedition reports, scientific datasets, publications, photographs, videos and institutional activities while generating content for websites and social media.”

What the statement requires, in plain English

Statement phrase	What it means for us	Our feature
Comprehensive outreach portal	A user-facing web platform, not a PDF-only archive.	PolarSetu web application
Expedition reports	Research/mission reports need cataloguing and discovery.	Reports library + expedition pages
Scientific datasets	Users need data records, metadata and links/downloads.	Dataset catalog + metadata cards
Publications	Research papers/publications need searchable metadata.	Publication library + filters
Photographs	Polar media needs structured discovery.	Media gallery + source metadata
Videos	Videos need organised access and context.	Video library + external embeds/links
Institutional activities	Outreach/events/announcements should be discoverable.	Activities/news timeline
Generating content for websites/social media	The platform should help turn source material into publication-ready drafts.	Polar Outreach Studio + approval queue

What the statement does NOT explicitly require

- A custom machine-learning model.
- A live weather prediction model.
- A 3D digital twin.
- Automatic social-media posting.
- A blockchain or Web3 component.
- A mobile app.
- A separate vector database.
- A massive web crawl or complete migration of all NCPOR data within the hackathon.

Judge-safe positioning

Stay faithful to the stated ask. Innovation should be shown through discovery, evidence, relationships, usability and dissemination—not through unrelated technology.

2. Research Dossier — What Already Exists

This section is crucial. The public NCPOR ecosystem already contains several relevant systems. That means our solution must not claim that polar information “does not exist.” The defensible problem is integration, discoverability, context, reuse and outreach experience.

2.1 NCPOR Library

NCPOR states that its library is a comprehensive polar library and that books, e-books, journals, news, Antarctic expedition reports and NCPOR publications are available through its online library. This is direct evidence that the repository problem is not “no digital archive exists.”

2.2 National Polar Data Center (NPDC)

The current NPDC exposes data availability, data search by free text, keyword/location browsing, station weather, technical reports, data submission, analytics and links to the Polar Directory. Its public page also groups datasets by science keywords such as atmosphere, cryosphere, oceans, paleoclimate and others.

2.3 Polar Directory

The Indian Antarctic Expedition Directory supports searches across institute, project/technical report, member/contributor, expedition, role, leader, keyword and station. This provides structured expedition/project metadata that can inform our model and relationships.

2.4 Antarctic records and publications

NCPOR's Antarctic Records page points to published technical reports and publication lists. This is a useful seed source for our demonstration corpus.

2.5 Outreach

NCPOR documents outreach activities including school visits, interactions with students, science fairs and scientific film festival participation. This directly supports the outreach angle of the problem statement.

2.6 Live/operational station data

The current NCPOR data site exposes station information for Maitri, Bharati, Himadri and Himansh; the live Maitri page shows station observations such as temperature, relative humidity, pressure, and wind. This can be treated as a future integration target or a read-only outbound link in the prototype rather than a fragile live-scraping dependency.

Research conclusion

The real product opportunity

NCPOR already has specialist information systems. Our product should sit above them as a unified discovery, evidence and dissemination experience: one place to find related resources, understand them, and create reviewable public-facing content.

2.7 Existing public SIH-style projects — what not to clone

A public GitHub project for SIH26063 called PolaRis already showcases an interactive globe, AI chatbot, dashboard, learning modules and media library. Its README describes a frontend-first architecture and lists backend migration, live telemetry, advanced RAG and role-based auth as future roadmap items.

A separate public PolarNCPOR demo showcases auto-generation of web/social content, image tagging, scheduling and engagement analytics. Whether or not it represents a final SIH submission, it is a useful competitive signal: those features should not be our only differentiators.

A generic public RAG foundation project demonstrates document ingestion, pgvector retrieval, hybrid search and source citations. We can learn the pattern, but our polar-specific data model and evidence chain should be original.

Competitive design rule

Borrow architectural lessons; do not copy UI, code, text, branding or dataset packaging. Our distinctive core is resource relationships + evidence lineage + audience-aware outreach + editorial control.

3. Product Vision — PolarSetu

Working name: PolarSetu. Meaning: a bridge between polar research, knowledge and the public. This is a project codename; check name availability before any public branding.

One-line pitch: “PolarSetu connects polar research resources into one searchable evidence trail and turns approved scientific knowledge into audience-specific outreach.”

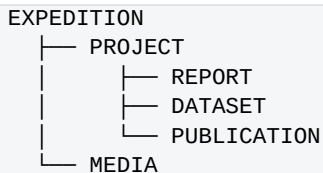
Tagline: “Discover the science. Understand the evidence. Share the story.”

3.1 Core pillars

Pillar	Purpose	What user sees
Discover	Find resources across the polar corpus.	Universal search, filters, map, timeline
Understand	Get context without losing source traceability.	Research pages, evidence-linked AI answers
Connect	Relate expeditions, projects, reports, datasets, publications and media.	Resource relationship panel
Create	Turn source material into audience-specific drafts.	Polar Outreach Studio
Review	Keep science communication controlled.	Approval queue + audit trail
Share	Publish approved content in the right format.	Web/social-ready exports

3.2 The “resource relationship” idea

Instead of treating every file as an isolated record, store explicit relationships: expedition → project → report → dataset → publication → media. This is a relational implementation of a knowledge graph idea without introducing Neo4j.



Any resource can show “Related evidence” and “Related media”.

3.3 Product loop

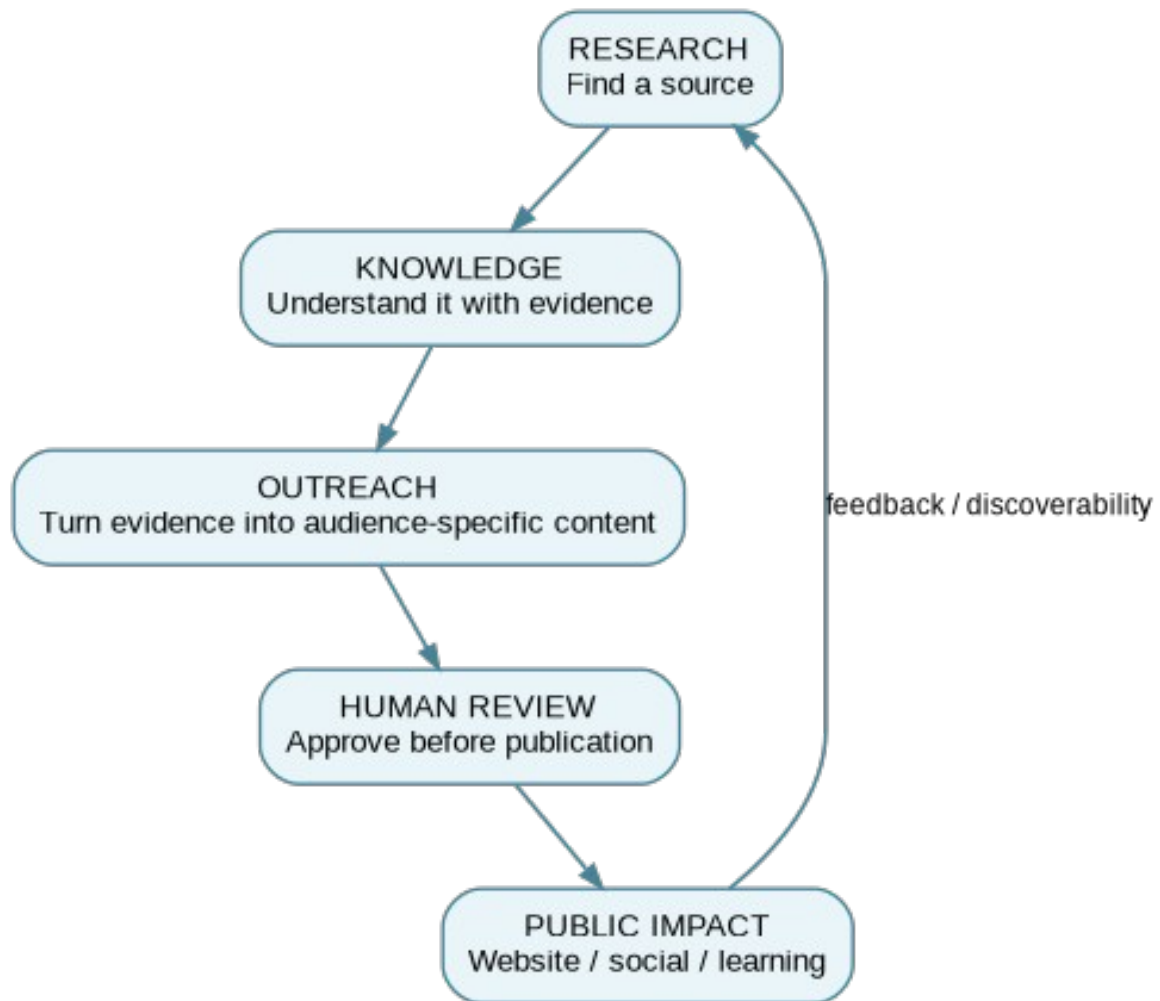


Figure: product loop — the repository supports discovery, AI understanding and controlled outreach.

4. Users, Personas and Jobs-to-be-Done

Persona	Need	Top screens	Example task
Researcher	Find related evidence quickly.	Search, resource detail, evidence panel	Find reports and datasets linked to an expedition.
Student	Understand science without reading a 100-page report.	Learn, AI assistant, media	Ask a simple question and open the sources.
Teacher / educator	Turn research into classroom-friendly material.	Outreach Studio, media	Create a student explanation and short script.
Public visitor	Explore India's polar presence.	Home, map, timeline, stories	Browse expeditions and photo stories.
NCPOR/portal admin	Curate and disseminate authoritative content.	Admin, repository, review queue	Approve a website article generated from a report.

Jobs-to-be-done

- When I know a topic but not the document, I want to search across resource types so I can find relevant evidence.
- When I open an expedition, I want to see the connected projects, reports, datasets and publications rather than searching separately.
- When I ask an AI question, I want the answer to show the sources used.
- When I have a scientific finding, I want to turn it into a student explanation, article or social draft without rewriting it from scratch.
- When AI generates content for an official channel, I want a human approval step before publishing.

5. Feature Scope — MVP, Stretch and Cut List

5.1 MVP — must be working

Feature	MVP definition	Demo priority
Home / landing	Value proposition + entry points to research, expeditions, media	High
Repository	Resource cards with type, year, topic, source and linked expedition	High
Search	Free-text search + type/year/topic filters	High
Resource detail	Metadata, source link/file, related resources	High
Expedition page	Overview + timeline + linked evidence	High
Map	Key station/expedition markers and popup cards	High
AI assistant	Source-grounded Q&A over seeded content	High
Outreach Studio	Generate article/social/student explanation/video script	High
Review queue	Draft → review → approve/reject	High
Admin	Add/edit catalog metadata and upload/link files	Medium

5.2 Stretch — only after MVP is reliable

- Hybrid search using pgvector + keyword search.
- Automatic topic extraction or media auto-tagging.
- Live NCPOR station data connector.
- Analytics dashboard for searches/content generation/approval.
- Multilingual outreach (Hindi + English first).
- Scheduled publishing export calendar.
- Embeddable “Polar Story” widgets for external websites.

5.3 Deliberately cut

- 3D polar digital twin.
- Custom AI model training.
- Real-time social posting to multiple platforms.
- OCR/image understanding for every upload.
- Neo4j knowledge graph.
- Mobile app.
- Complex notification infrastructure.
- Live satellite ingestion.
- Automatic scientific fact-checking beyond source checks.

6. Screen-by-Screen Product Blueprint

6.1 Home

- Hero: “India’s polar science, connected.”
- Global search box: “Search reports, datasets, publications, expeditions...”
- Quick access: Expeditions | Research | Datasets | Media | Ask PolarSetu
- Featured expedition or current story
- Small map preview
- Impact strip: number of catalogued resources / expeditions / research topics

6.2 Repository

- Search field with debounce
- Filters: resource type, region, year, expedition, science topic
- Card includes title, year, type, source, tags, related expedition
- Sort: relevance, newest, oldest

6.3 Resource detail

- Title, abstract/description
- Metadata table
- Source link / file viewer
- Related expedition
- Related projects
- Related datasets/publications/media
- “Ask about this source” button

6.4 Expedition detail

- Location and date
- Objective
- Research areas
- Map marker
- Timeline
- People/projects if public
- Linked reports/datasets/publications/media
- “Generate outreach pack” button

6.5 Ask PolarSetu

- Question box
- Answer
- Source cards with source title + section/page when known
- “Open source” link
- Notice when evidence is insufficient

6.6 Polar Outreach Studio

- Choose source(s)
- Choose audience: researcher / student / general public
- Choose output: article / social / press note / student explainer / video script
- Preview structured draft
- Source list
- Approve / reject / regenerate

6.7 Admin review

- Draft queue
- Source provenance
- Generated time

- Editor comments
- Approve / reject
- Audit trail

6.8 Map

Use Leaflet with station/expedition markers. Show only a small, controlled set in MVP. Use proper map attribution if OpenStreetMap tiles are used.

7. System Architecture

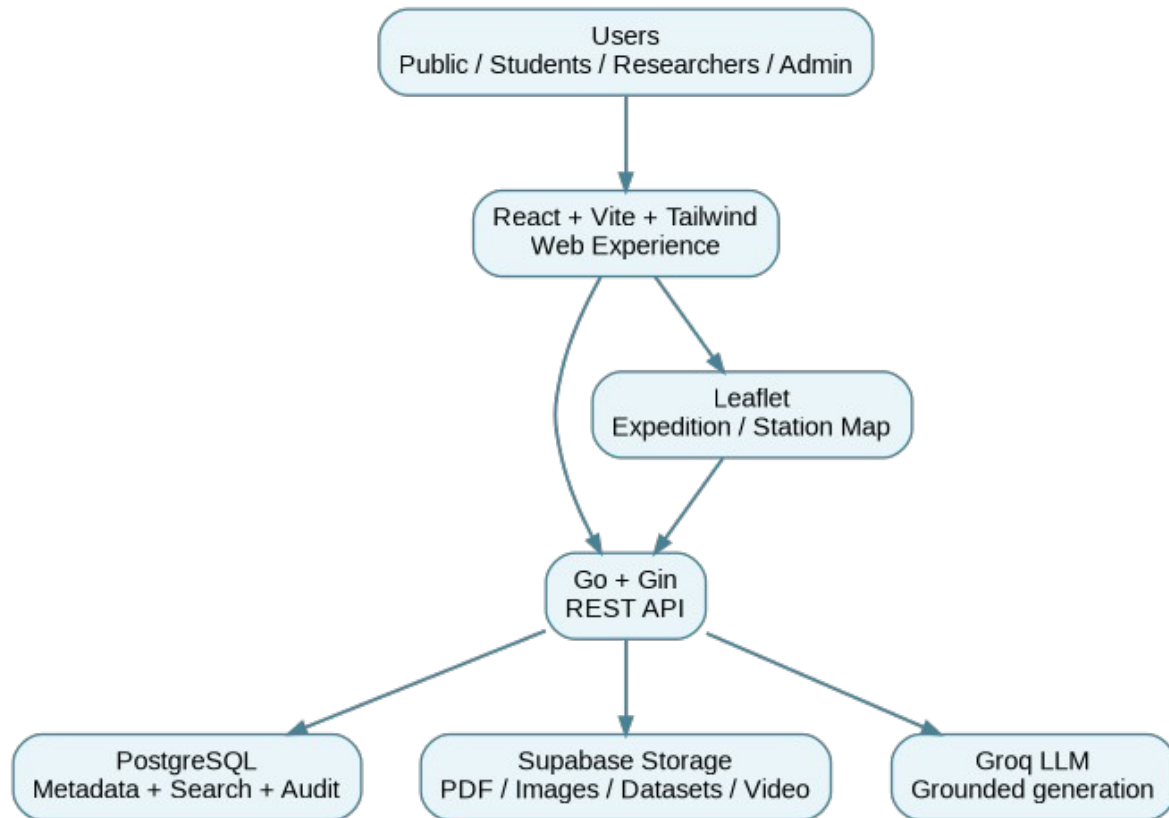


Figure: recommended 3-day architecture. One frontend, one Go backend, one database, one storage layer and one LLM provider.

7.1 Why this architecture

- Keeps the backend in Go as requested.
- Avoids a second backend language/service for AI.
- Uses PostgreSQL for both application data and search, reducing infrastructure.
- Separates object storage from metadata.
- Keeps AI behind the server so keys are not exposed.
- Makes each layer replaceable later without rewriting the product concept.

7.2 Production-minded evolution

MVP

React → Go/Gin → PostgreSQL + Storage → Groq

Next phase

React → Go/Gin → PostgreSQL + pgvector → Groq

Scale phase

Connectors → ingestion jobs → search index → moderation/audit → analytics

8. Final Technology Stack — Locked

Layer	Technology	Why	Use now?
Frontend	React + Vite	Fast UI development and component reuse	Yes
Styling	Tailwind CSS	Rapid responsive design	Yes
Routing	React Router	Simple page navigation	Yes
Icons	Lucide React	Consistent lightweight icons	Yes
Charts	Recharts	Simple dashboard visuals	Optional
Map	Leaflet + React Leaflet	Interactive station/expedition map	Yes
Backend	Go	Requested backend language; simple deployment	Yes
HTTP framework	Gin	Fast REST API prototyping and middleware	Yes
PostgreSQL driver	pgx/v5	Native Go PostgreSQL driver/toolkit	Yes
Database	PostgreSQL	Structured data + search	Yes
Text search	PostgreSQL FTS + pg_trgm	No separate search cluster	Yes
Vector search	pgvector	Semantic retrieval in PostgreSQL	Later
File storage	Supabase Storage	Documents/media without storing blobs in DB	Yes
Authentication	JWT + bcrypt	Admin-only login for MVP	Yes, if needed
AI	Groq API	Fast hosted LLM, structured output support	Yes
HTTP AI client	Go net/http	Avoid another SDK dependency	Yes
PDF text extraction	ledongthuc/pdf (prototype)	Simple Go text extraction for text-based PDFs	Optional / prototype only
API docs	Swagger/OpenAPI	Easy endpoint demo and team coordination	Optional
Version control	Git + GitHub	Collaboration + submission artifact	Yes
Testing	Go tests + Postman	Core API checks	Yes
Containerization	Docker Compose	Reproducible local environment	Optional after MVP
Frontend hosting	Vercel	Fast static deployment	Optional
Backend hosting	Go-friendly host such as Render	Simple deployment	Optional

Evidence from current public documentation

Gin is designed for REST APIs and web applications and currently documents middleware, JSON binding/validation and routing. pgx is a native Go PostgreSQL driver/toolkit. Supabase Storage supports file upload/download and access control. Groq supports structured JSON-schema outputs for selected models. Leaflet provides interactive maps and GeoJSON support.

Implementation note: vendor features, model availability, hosting limits and free-tier details can change; verify the exact service/model before final deployment.

9. Data Model and Database Design

9.1 Recommended tables

Table	Core fields	Notes
users	id, email, password_hash, role, created_at	Admin only for MVP
expeditions	id, name, region, year, start_date, end_date, objective, latitude, longitude, source_url	Parent object
resources	id, type, title, description, year, region, source_url, storage_path, license, status, created_at	Shared catalog table
resource_expedition	resource_id, expedition_id	Many-to-many relationship
resource_relations	from_resource_id, to_resource_id, relation_type	Related evidence graph without Neo4j
resource_chunks	id, resource_id, page_number, section, content, search_vector	For source-grounded Q&A
activities	id, title, date, description, source_url, media_url	Institutional activity timeline
media	id, resource_id, media_type, url, caption, attribution	Photo/video metadata
ai_generations	id, user_id, source_ids, audience, output_type, content, status, created_at	Draft/audit trail
audit_log	id, actor_id, action, entity_type, entity_id, created_at	Approval/history

9.2 Resource types

REPORT
PUBLICATION
DATASET
PHOTO
VIDEO
ACTIVITY
EXPEDITION
OTHER

9.3 Resource status

DRAFT → IN_REVIEW → APPROVED → PUBLISHED
 REJECTED

9.4 Search design

PostgreSQL Full-Text Search can index a tsvector built from title, description, tags and extracted text. PostgreSQL documentation recommends GIN as the preferred index type for text search. pg_trgm is useful as a complementary similarity/fuzzy matching mechanism.

```
-- conceptual example
CREATE INDEX resources_search_idx
ON resources USING GIN (search_vector);

-- later: use pg_trgm on title / keywords for fuzzy matching
```

9.5 Why relational relationships instead of Neo4j

Our relationship graph is small and structured: expedition-to-report, report-to-dataset, publication-to-project, media-to-expedition. PostgreSQL handles these joins cleanly. Neo4j adds setup, deployment and query complexity without being necessary for the 72-hour prototype.

10. Metadata, Provenance and FAIR Principles

The FAIR principles aim to make digital assets Findable, Accessible, Interoperable and Reusable. We will not claim full FAIR compliance in a 3-day prototype; instead, we adopt the principles as design guidance.

Principle	Our implementation
Findable	Unique resource ID, searchable metadata, tags, filters and persistent source URL.
Accessible	Clear source link or approved storage path; metadata visible even if the original file is not mirrored.
Interoperable	Stable JSON API, normalized resource types, ISO-like dates, standard geographic fields.
Reusable	License/rights field, attribution, description, provenance and related-resource relationships.

10.1 Minimum provenance fields

```
source_id
source_url
source_title
source_type
source_date
page_or_section
license_or_rights
retrieved_at
checksum_sha256
```

Critical scientific rule

AI output must never be presented as the primary source. The original source remains authoritative. AI creates an interpretation/draft layer on top of retrieved evidence.

11. End-to-End Data Ingestion Workflow



11.1 MVP ingestion path

33. Admin selects “Add Resource”.
34. Enter metadata: title, type, year, region, expedition, source URL, license/rights, description.
35. Optional: upload PDF/image/CSV/video or store external link.
36. Go computes SHA-256 for duplicate detection.
37. If a text PDF is provided and extraction succeeds, store extracted text and page boundaries.
38. Generate search_vector from metadata + text.
39. Resource becomes searchable.
40. AI can use only approved/indexed resource text as context.

11.2 Production ingestion path

For a production deployment, source connectors can periodically ingest metadata from NCPOR Library, NPDC, PolarDirectory and other approved feeds/APIs. Store source IDs and URLs rather than blindly copying the entire corpus.

11.3 File strategy

- Keep large binaries out of PostgreSQL.
- Store files in object storage.
- Store only metadata and paths/URLs in PostgreSQL.
- For media you do not own, prefer link-out/embedded viewing with attribution instead of re-hosting.

12. AI Architecture — Grounded, Structured, Controlled

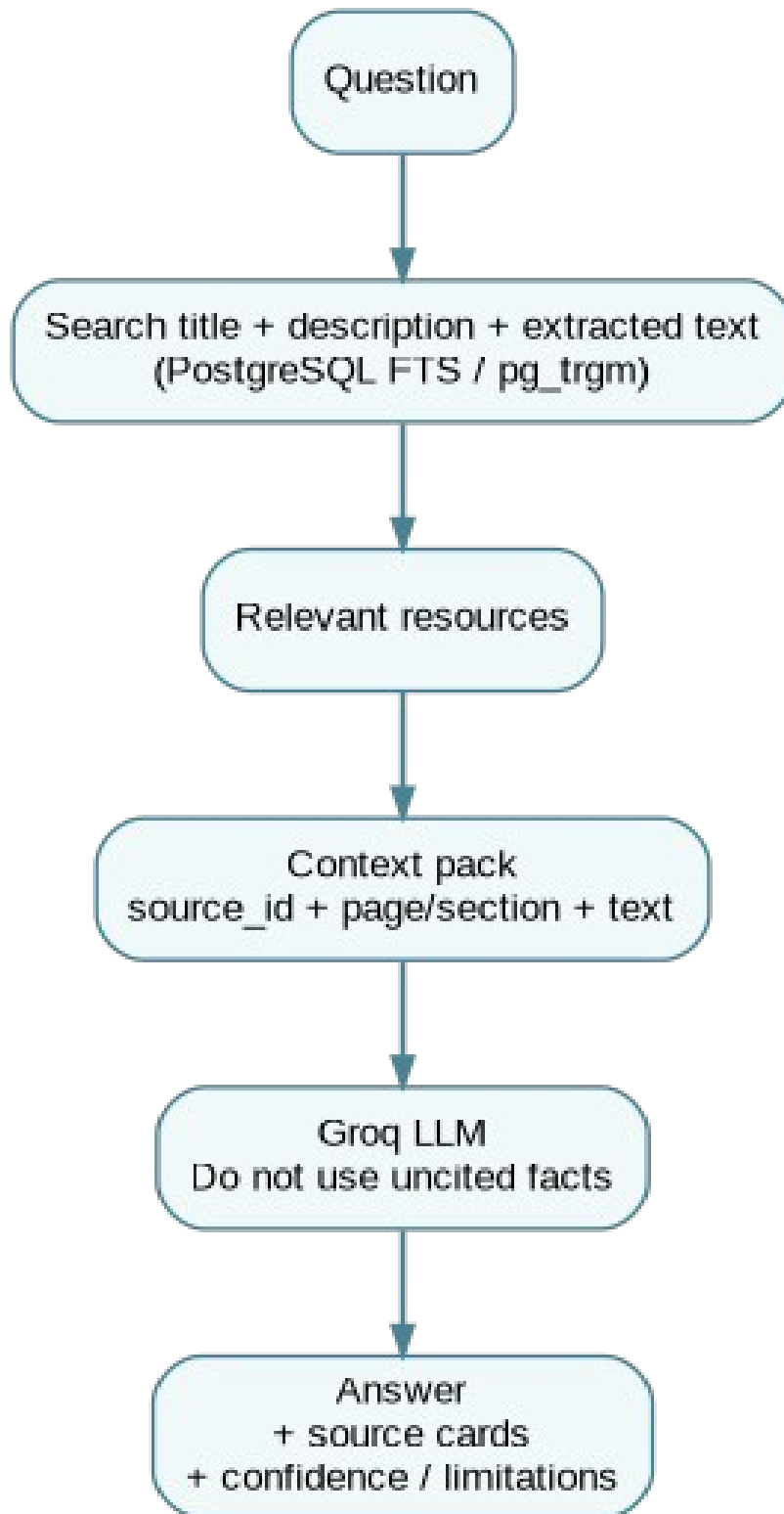


Figure: keyword-grounded RAG for the MVP. pgvector is a later enhancement, not a dependency for the first working demo.

12.1 AI features

Feature	Input	Output	Grounding
Ask PolarSetu	Question + optional resource/expedition context	Answer + source cards	Required
Summarize	Selected report/resource	Structured summary	Required

Feature	Input	Output	Grounding
Student explain	Selected finding	Simple explanation	Required
Website article	Selected evidence	Article draft	Required
Social post	Selected evidence	Platform-length draft	Required
Video script	Selected evidence	Short script	Required

12.2 AI rules

- Use only retrieved context for factual claims about the source corpus.
- If context is insufficient, say “The available sources do not provide enough evidence.”
- Never invent citations, page numbers, authors or measurements.
- Preserve uncertainty present in the source.
- Prefer concise source cards over invisible citations.
- Separate source facts from generated interpretation.
- Generated official-channel content is always a draft until approved.

12.3 Structured output

```
{
  "title": "...",
  "summary": "...",
  "key_points": [...],
  "audience": "student",
  "content_type": "website_article",
  "source_ids": ["RPT-2025-001", "PUB-2025-014"],
  "warnings": [...],
  "status": "draft"
}
```

Groq currently documents structured outputs using JSON Schema for supported models. Pin the model in configuration and verify current model support before deployment. For the hackathon, using structured output reduces frontend/backend parsing problems.

12.4 Suggested system prompt

You are PolarSetu, a source-grounded polar-science assistant.

Rules:

1. Use only the supplied evidence context for factual claims.
2. Never fabricate authors, dates, statistics, citations or page numbers.
3. When evidence is insufficient, say so explicitly.
4. Keep scientific uncertainty intact.
5. Return the requested JSON schema exactly.
6. Every factual answer must include the supplied source IDs.
7. You are generating drafts, not publishing official content.

13. Polar Outreach Studio

The outreach studio is the main feature that connects the AI layer directly to the problem statement. The important design decision is to generate multiple audience/channel formats from the same verified source set.

Audience	Tone	Example output
Researcher	Precise, technical, source-rich	Research summary with source links
Student	Simple, educational	"Explain this finding in class-9 language"
General public	Accessible, engaging, accurate	Short article or social post

13.1 Content workflow

41. Select resource/expedition.
42. System displays the sources selected.
43. Choose audience and output format.
44. Go retrieves the relevant source context.
45. Groq generates structured draft.
46. System stores draft + source IDs.
47. Editor reviews/edits.
48. Approve or reject.
49. Approved version is exportable as text/Markdown/HTML.

13.2 No auto-publishing in MVP

Why

Automatic posting creates integration, authentication and governance complexity. The MVP should demonstrate "publication-ready output + review workflow". Real social APIs can be future scope.

13.3 Example demo

```

SOURCE
15th Indian Arctic Expedition Report (2024-2025)
  ↓
SELECT: Student Explanation
  ↓
AI DRAFT
What scientists studied...
Why it matters...
  ↓
SOURCES
Report, page/section
  ↓
EDITOR REVIEW
[Approve] [Reject] [Edit]
```

14. Backend API Contract

All application logic goes through Go/Gin REST endpoints. Keep the API surface small and predictable.

Method	Endpoint	Purpose
GET	/api/health	Health check
POST	/api/auth/login	Admin login
GET	/api/expeditions	List expeditions
GET	/api/expeditions/:id	Expedition detail
POST	/api/expeditions	Create expedition
GET	/api/resources	List/filter resources
GET	/api/resources/:id	Resource detail
POST	/api/resources	Create resource
POST	/api/resources/:id/upload	Attach file
GET	/api/search?q=	Search
GET	/api/media	Media list
GET	/api/activities	Activities/news
POST	/api/ai/ask	Grounded Q&A
POST	/api/ai/summarize	Summary draft
POST	/api/ai/outreach	Outreach generation
GET	/api/review/queue	Review queue
POST	/api/review/:id/approve	Approve draft
POST	/api/review/:id/reject	Reject draft

14.1 Example search request/response

GET /api/search?q=sea%20ice&type=REPORT,PUBLICATION&year=2025

```
{
  "query": "sea ice",
  "results": [
    {
      "id": "RPT-2025-001",
      "title": "...",
      "type": "REPORT",
      "year": 2025,
      "expedition": "...",
      "score": 0.92
    }
  ]
}
```

14.2 Example AI request

POST /api/ai/ask

```
{
  "question": "What did this expedition study?",
  "resource_ids": ["RPT-2025-001"]
}
```

14.3 Error contract

```
{
  "error": {
    "code": "RESOURCE_NOT_FOUND",
    "message": "Requested resource was not found"
  }
}
```

15. Go Backend Structure

```
backend/
├── cmd/server/main.go
├── internal/
│   ├── config/
│   ├── db/
│   ├── models/
│   ├── repository/
│   ├── handlers/
│   ├── services/
│   │   ├── search.go
│   │   ├── ai.go
│   │   ├── storage.go
│   │   └── provenance.go
│   ├── middleware/
│   └── routes/
├── migrations/
├── docs/
└── go.mod

frontend/
├── src/
│   ├── components/
│   ├── pages/
│   ├── services/
│   ├── hooks/
│   ├── types/
│   └── App.jsx
└── package.json
```

Recommended Go dependencies

```
go get github.com/gin-gonic/gin
go get github.com/jackc/pgx/v5
go get github.com/golang-jwt/jwt/v5
go get golang.org/x/crypto/bcrypt
go get github.com/joho/godotenv
# Optional prototype PDF extraction:
go get github.com/ledongthuc/pdf
```

Use the latest compatible versions at build time. Current Gin/pgx documentation should be consulted because requirements can change.

16. Frontend Structure

```
frontend/src/
├── components/
│   ├── Navbar.jsx
│   ├── SearchBar.jsx
│   ├── ResourceCard.jsx
│   ├── FilterPanel.jsx
│   ├── SourceCard.jsx
│   ├── ExpeditionMap.jsx
│   └── StatusBadge.jsx
├── pages/
│   ├── Home.jsx
│   ├── Repository.jsx
│   ├── ResourceDetail.jsx
│   ├── ExpeditionDetail.jsx
│   ├── Ask.jsx
│   ├── OutreachStudio.jsx
│   ├── ReviewQueue.jsx
│   └── Admin.jsx
├── services/
│   └── api.js
├── hooks/
├── types/
└── App.jsx
```

UI rules

- Do not use a generic admin-template look.
- Use large scientific imagery sparingly and only with rights/attribution.
- Use consistent card hierarchy and labels.
- Every AI answer gets a visible source area.
- Every official-channel draft gets a visible “DRAFT — human review required” state.
- Keep navigation understandable in 5 seconds.

17. Expedition Map and Geographic Layer

Use Leaflet + React Leaflet. Leaflet is intentionally provider-agnostic; if using OpenStreetMap tiles, follow the tile usage policy and show visible attribution.

MVP map content

- Indian Antarctic research stations relevant to the story
- Indian Arctic research station
- Himalayan polar-research station
- Expedition markers (optional)
- Click marker → summary card → “Open expedition”

Do not

- Claim the coordinates are official unless sourced.
- Scrape or bulk-download map tiles.
- Use OSM tiles without attribution.
- Build route planning or navigation unless explicitly needed.

Map data model

```
station_name
region
latitude
longitude
station_type
source_url
last_verified_at
```

18. Research/Data Sources for the Demo

The first demo corpus should be small, real and traceable. Do not attempt to ingest the entire polar data universe in three days.

Source	Use in demo	Acquisition method
NCPOR Antarctic Records	Technical reports/publication references	Link or manually seed metadata
NCPOR Arctic reports	Real expedition report example	Download official report only where permitted; otherwise link
NPDC datasets	Dataset cards and keywords	Seed metadata + official link
PolarDirectory	Expedition/project relationship examples	Seed metadata from public records
NCPOR live station page	Optional live-data link / snapshot	Outbound link; avoid brittle scraping
NCPOR outreach page	Activity examples	Seed activity metadata + source URL
Semantic Scholar API	Optional publication discovery	Search API, cache only what you need
Wikimedia Commons / other properly licensed sources	Optional public-domain/CC media	Use only assets with compatible licences and record attribution

18.1 Suggested seed corpus

Type	Target count	Example
Expeditions	4–6	A few Antarctic + Arctic entries
Reports	6–10	Official expedition/technical reports
Publications	8–15	NCPOR publication metadata + selected linked papers
Datasets	8–12	NPDC metadata cards across science keywords
Activities	6–10	Outreach/news items
Media	10–20	Only properly sourced/licensed or linked assets

Demo-data principle

A smaller corpus with excellent metadata and clear relationships is better than hundreds of fake rows.

19. Security, Privacy and Governance

Area	Minimum control
API secrets	Store GROQ_API_KEY and DB secrets only in server environment variables.
Frontend	Never expose secret API keys in VITE_* variables.
Admin auth	JWT + bcrypt, short expiry, protected routes.
Uploads	Validate MIME type, extension and max size; generate safe object names.
Duplicate detection	Compute SHA-256 before indexing.
Search	Use parameterized SQL; never concatenate raw user input into SQL.
AI endpoints	Rate-limit and log usage; reject empty/huge prompts.
Audit	Record who created/edited/approved official content.
Source rights	Record license/attribution/official source URL.
Deployment	Disable debug mode; configure CORS explicitly.
Repository hygiene	Keep .env out of Git, use secret scanning and review commits before sharing.

19.1 Copyright warning

NCPOR's current copyright policy page states that the site content is copyrighted/all rights reserved. The safest hackathon pattern is therefore: use official pages and links as evidence; re-host only content that the team has clear permission/license to use; and store attribution/rights metadata.

20. Testing and Demo Reliability

20.1 Smoke tests

50. Open home page.
51. Search “sea ice” and verify results.
52. Open one expedition.
53. Open one resource.
54. Ask one grounded AI question and verify sources appear.
55. Generate one social post and verify it is marked Draft.
56. Approve the draft and verify status changes.
57. Refresh page and verify database-backed state persists.
58. Open map and click a marker.
59. Test API health endpoint.

20.2 AI evaluation sample

Test	Expected
Question fully covered by a report	Answer uses report and cites it.
Question outside corpus	System says evidence is insufficient.
Conflicting sources	System surfaces both and avoids pretending one is certain.
No resource selected	Ask endpoint returns a clear validation response.
Malformed AI output	Server rejects/handles safely rather than saving bad JSON.
Re-run generation	New draft has new generation ID and preserves old audit record.

21. 72-Hour Execution Plan

Treat these as three consecutive phases. Never start a lower-priority task while a higher-priority phase is unstable.

Hour 0–6: Foundation

- Create GitHub repo and branches.
- Install Go, Node.js, PostgreSQL/Supabase access.
- Scaffold React/Vite + Tailwind.
- Scaffold Go/Gin server.
- Create database connection and health endpoint.

Hour 6–18: Core backend + database

- Create migrations/tables.
- Seed expeditions/resources.
- Implement CRUD for resources/expeditions.
- Implement search.
- Add upload/storage metadata.

Hour 18–30: Core frontend

- Home.
- Repository list.
- Filters/search.
- Resource detail.
- Expedition detail.
- Basic map.

Hour 30–42: AI

- Grounded Q&A.
- Source cards.
- Summary.
- Outreach Studio.
- Structured output.

Hour 42–54: Review + provenance

- Draft/review/approve flow.
- Audit log.
- Source IDs/page references.
- Error states.

Hour 54–66: Polish

- Seed better data.
- Improve visual consistency.
- Fix responsiveness.
- Add demo story.
- Write README.

Hour 66–72: Freeze

- No new features.
- Run smoke tests.
- Deploy.
- Record screenshots/video.
- Finalize PPT and speaking script.
- Tag a release in Git.

21.1 Team allocation

Role	Primary responsibility	Backup
Team Lead / Product	Scope, integration, demo narrative	Any backend member
Go Backend	API, DB, auth, AI service	Product lead
Frontend	React pages, components, map	UI/data member
Data + AI	Seed corpus, retrieval, prompts, content quality	Backend
Research + PPT	Source verification, references, slides	Product
QA + DevOps	Testing, deployment, README, screenshots	Everyone

Team rule

One person owns each subsystem. No shared “everyone edits everything” workflow during the final 24 hours.

22. SIH 6-Slide Idea Submission — Ready-to-Paste Content

The uploaded SIH template specifies a maximum of six slides including the title slide, recommends concise points/diagrams/infographics rather than paragraphs, says to keep the explanation precise and easy to understand, requires using the provided template, and says the upload should be a PDF.

Your uploaded template has these sections: Title Page; Idea Title; Technical Approach; Feasibility and Viability; Impact and Benefits; Research and References. Use those exact headings in the official template.

Slide 1 — TITLE PAGE

SMART INDIA HACKATHON 2026

Problem Statement ID - SIH26063

Problem Statement Title - Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal

Theme - Smart Education / Space Technology (use the label shown on the current official portal/template)

PS Category - Software

Team ID - <TEAM ID>

Team Name - <REGISTERED TEAM NAME>

Optional small footer: PolarSetu

Keep this slide visual and clean. No paragraphs.

Slide 2 — IDEA TITLE

IDEA TITLE

PolarSetu – Unified Discovery, Evidence & Outreach Layer for India's Polar Science

PROPOSED SOLUTION

- One searchable portal for reports, datasets, publications, photos, videos and institutional activities.
- Resource relationships connect expedition → project → report → dataset → publication → media.
- Evidence-linked AI assistant answers questions from the indexed source corpus.
- Polar Outreach Studio converts selected scientific evidence into website, social, student and video drafts.
- Human review is required before official publication.

HOW IT ADDRESSES THE PROBLEM

Discover → Understand → Create → Review → Share

INNOVATION / UNIQUENESS

- Unified discovery across existing specialist polar resources
- Provenance-first AI with visible source cards
- Audience-aware scientific communication
- Editorial control and audit trail

Slide 3 — TECHNICAL APPROACH

TECH STACK

React + Vite + Tailwind | Go + Gin | PostgreSQL + pg_trgm/FTS | Supabase Storage | Groq | Leaflet

ARCHITECTURE

User

↓

React UI

↓ REST/JSON

Go + Gin

├─ PostgreSQL (metadata + search + audit)

├─ Storage (documents/media)

— Groq (source-grounded generation)

METHODOLOGY

1. Catalog/ingest resource + metadata
2. Extract/index text where available
3. Search relevant sources
4. Build evidence context
5. Generate structured output
6. Attach source IDs
7. Human review
8. Export/publish

Slide 4 — FEASIBILITY AND VIABILITY

FEASIBILITY

- Single Go backend + single PostgreSQL database minimizes operational complexity.
- MVP uses keyword search; pgvector is optional later.
- Seeded, traceable demo corpus avoids dependence on large live integrations.
- Public NCPOR resources provide realistic source structure for prototyping.

RISKS

- Copyright / media rights
- AI hallucination
- Unstable live integrations
- Limited 72-hour development window

MITIGATION

- Store rights/attribution metadata; prefer links for third-party content
- Source-grounded prompts + visible citations + human review
- Use outbound official links before live connectors
- Freeze MVP early; no microservices / no ML training

Slide 5 — IMPACT AND BENEFITS

TARGET AUDIENCE

Researchers | Students | Educators | Public | Institutional communicators

IMPACT

- Faster discovery of polar research resources
- Better connection between expeditions and their evidence
- Scientific information becomes accessible to non-experts
- Faster production of outreach content
- Human-reviewed dissemination improves trust and control

BENEFITS

Social: science outreach and STEM engagement
 Educational: student-friendly explanations and media
 Scientific: clearer discovery and provenance
 Operational: reusable content workflow and metadata

Slide 6 — RESEARCH AND REFERENCES

KEY RESEARCH

- NCPOR Library – digital polar knowledge resources and expedition reports
- National Polar Data Center – datasets, search, technical reports and data services
- Indian Antarctic Expedition Directory – expedition/project metadata
- NCPOR Outreach Program – school/science outreach examples
- Official NCPOR Arctic/Antarctic reports and publication lists
- FAIR Principles – metadata and reusable scientific-data design
- PostgreSQL FTS / pg_trgm – low-complexity search layer
- Groq structured outputs – reliable JSON response format

```
SELECT REFERENCES
ncpor.res.in/libraries
npdc.ncpor.res.in
data.ncpor.res.in/PolarDirectory/home
...
```

For the real submission, place short source names and URLs in the slide, not a wall of raw URLs. Keep the detailed reference list in this handbook/repository.

23. Demo Script — What the Judge Should See

Target demo duration: about 4–5 minutes, adjusted to the actual presentation rules. Demonstrate one coherent story rather than every feature.

Opening (20–30 sec)

“Polar science produces reports, datasets, publications and media. NCPOR already operates specialist information systems for these resources. Our solution is the layer that connects discovery, evidence and outreach into one user experience.”

Scene 1 — Discover (40 sec)

Search for a real polar topic such as “sea ice”. Show filters and a result set. Open one resource and point to its metadata and official source.

Scene 2 — Connect (40 sec)

Open the expedition page. Show the map, timeline and related evidence. Explain that the resource relationship model connects the expedition to reports, datasets, publications and media.

Scene 3 — Understand (60 sec)

Ask: “What did this expedition study?” Show an answer with visible source cards. Then ask something outside the corpus and show the system refusing to invent a source-based answer.

Scene 4 — Create (60 sec)

Select a report → open Polar Outreach Studio → choose Student → Generate Student Explanation. Then generate a social or website draft from the same source set.

Scene 5 — Govern (40 sec)

Open review queue. Show DRAFT status and source IDs. Approve it. Explain that official content is not auto-published in the MVP.

Closing (20–30 sec)

“PolarSetu is not another isolated archive or chatbot. It is a source-aware discovery and dissemination layer that makes existing polar knowledge easier to find, understand, connect and responsibly share.”

24. Judge Q&A — Prepared Answers

Q: Why does this need a new portal when NCPOR already has a library and data portal?

A: We are not claiming those systems do not exist. Our research found NCPOR Library, NPDC, PolarDirectory and station-data interfaces. Our proposal is an integration and experience layer that connects those resource types, exposes related evidence, and adds controlled outreach workflows.

Q: Is this just a chatbot over PDFs?

A: No. The chatbot is one component. The main product is the resource graph/catalog, search, expedition experience, provenance layer and outreach workflow. The AI operates on top of that structured source layer.

Q: How do you reduce hallucinations?

A: Retrieve source context first, send only that context to the model, require source IDs in the structured output, and allow human review before official publication. When evidence is insufficient, the assistant should say so.

Q: Why Go?

A: Go keeps the backend single-language, lightweight and straightforward for REST APIs, concurrent requests and deployment. It also matches our team preference.

Q: Why PostgreSQL instead of Elasticsearch?

A: The MVP is small. PostgreSQL already provides full-text search and GIN indexes, so we avoid an extra service. Elasticsearch is a scale option, not a requirement for the prototype.

Q: Why not use Neo4j for the knowledge graph?

A: Our relationship model is small and structured. A relational table for resource relationships gives us graph-like navigation without adding another database and deployment dependency.

Q: How is this different from existing public SIH projects?

A: We deliberately avoid treating a globe or chatbot as the core novelty. Our differentiation is evidence lineage, resource-to-resource relationships, audience-aware outreach generation and an explicit review/audit step.

Q: Where do your documents come from?

A: We use a small, traceable demo corpus based on official NCPOR public pages/resources and properly licensed media. The production architecture is designed around source connectors and metadata rather than unrestricted scraping.

Q: Will you automatically publish to social media?

A: Not in the 72-hour MVP. We generate platform-ready drafts and provide a review/approval workflow. Direct API-based publishing is a future integration.

Q: Is the AI scientifically trustworthy?

A: The model itself is not the authority. The source document is. Our UI makes that explicit, shows the source trail, and prevents unreviewed official publication.

Q: How will it scale?

A: The API is stateless, PostgreSQL can scale vertically, storage is externalized, and source connectors can be added independently. Search can later use pgvector or a dedicated search engine when corpus size requires it.

Q: What is the measurable benefit?

A: We can measure search success, time-to-source, AI response source coverage, draft creation time, approval time, and reuse of a source across multiple outreach formats. These become operational metrics after deployment.

25. Future Roadmap

Phase	Capability	Value
Phase 1 — MVP	Unified catalog, search, relationships, AI assistant, outreach review	Demonstrable solution
Phase 2	pgvector/hybrid retrieval, multilingual support, better document ingestion	Better discovery and accessibility
Phase 3	Connectors to approved NCPOR systems, live station data, automated synchronization	Operational integration
Phase 4	Multimodal search, image tagging, media understanding, richer analytics	Scale of media and knowledge
Phase 5	Federated scientific repository, advanced provenance, publication intelligence	Long-term institutional platform

What future scope should NOT be used to hide MVP gaps

- Do not call an unimplemented feature “working”.
- Do not show screenshots of future scope as proof of implementation.
- Use a clear “implemented / demo / future” label in the final presentation.

26. GitHub Repository and Engineering Hygiene

```
polarsetu/
├── README.md
├── LICENSE
├── .gitignore
├── .env.example
├── docker-compose.yml      # optional after MVP
├── backend/
├── frontend/
├── migrations/
├── seed/
├── docs/
│   ├── architecture.md
│   ├── api.md
│   ├── data-sources.md
│   └── demo-script.md
└── screenshots/
```

README must contain

- Problem statement summary
- Solution summary
- Architecture diagram
- Tech stack
- Local setup
- Environment variables
- Database migration/seed steps
- Demo credentials if applicable
- API overview
- Known limitations
- Future roadmap
- Research references
- Attribution / rights notes

Git workflow

```
main
├── feat/backend
├── feat/frontend
├── feat/ai
├── feat/data
└── feat/ppt
```

PR → review → merge
No force-push to main in final 24 hours.

27. Day-1 Setup Commands

27.1 Frontend

```
npm create vite@latest frontend -- --template react
cd frontend
npm install
npm install react-router-dom lucide-react axios leaflet react-leaflet
# Tailwind installation command depends on the current Tailwind version; follow the current
official guide.
```

27.2 Backend

```
mkdir backend
cd backend
go mod init github.com/<org>/<repo>/backend
go get github.com/gin-gonic/gin
go get github.com/jackc/pgx/v5
go get github.com/golang-jwt/jwt/v5
go get golang.org/x/crypto/bcrypt
go get github.com/joho/godotenv
go run ./cmd/server
```

27.3 Environment variables

```
DATABASE_URL=postgresql://...
GROQ_API_KEY=...
JWT_SECRET=...
SUPABASE_URL=...
SUPABASE_SERVICE_KEY=...
MAX_UPLOAD_MB=25
APP_ENV=development
```

Secret rule

The Supabase service key and Groq key stay server-side. The React application receives only public configuration and authenticated API responses.

28. Common Failure Modes and Recovery Plan

Failure	First response	Fallback
DB connection fails	Check DATABASE_URL and network	Use local PostgreSQL for dev
AI API fails	Show graceful error + retry	Demo with a pre-generated response from the same approved source; clearly label demo fallback
PDF parser fails	Do not block core repository	Use pre-extracted demo text or a stored source excerpt
Map tiles fail	Show static map section / outbound link	Keep map markers and expedition cards; restore tiles later
Deployment fails	Run local demo on laptop	Use screen-recording/screenshots only as last resort; do not claim live deployment
Frontend build breaks	Revert latest feature commit	Use stable tagged build
Search returns poor results	Simplify to title/description/tag matching	Use curated synonym tags

Golden rule for the last 12 hours

Freeze

Do not add features. Fix reliability, polish the demo, validate the PPT, and rehearse the story.

29. Research and Reference Library

The following sources were used to shape this handbook. The list intentionally separates official/current institutional sources from community/GitHub reference implementations and technical documentation.

Source	Why it matters	URL	Notes
SIH26063 current public problem listing	Problem title, organization, category and exact statement.	https://sih2026.vuce.in/ps/SIH26063	Community-maintained SIH 2026 archive; linked to official portal.
NCPOR Library	Existing digital polar library, expedition reports, publications and preservation/dissemination mission.	https://ncpor.res.in/libraries	Official NCPOR site.
National Polar Data Center	Data search, datasets, technical reports, station data, data submission and analysis interfaces.	https://npdc.ncpor.res.in/	Official NCPOR/NPDC site.
NCPOR Polar Directory	Searchable expedition/project/member/station metadata.	https://data.ncpor.res.in/PolarDirectory/home	Official NCPOR data service.
NCPOR Antarctic Records	Technical reports and publication references.	https://ncpor.res.in/pages/display/384-antarctic-records	Official NCPOR site.
15th Indian Arctic Expedition Report	Real public report useful for demo source structure.	https://www.ncpor.res.in/arctics/display/452-reports	Official NCPOR site.
NCPOR Outreach Program	Evidence of science and school outreach activity.	https://ncpor.res.in/pages/display/373-outreachprogram	Official NCPOR site.
NCPOR live Maitri data	Example live station observations.	https://data.ncpor.res.in/live/maitri	Official NCPOR data service.
NCPOR Copyright Policy	Rights/usage caution for content on NCPOR site.	https://ncpor.res.in/pages/display/33-copyright-policy	Official NCPOR site.
National Polar Data Center User Manual	Background on NPDC role and polar data management.	https://npdc.ncpor.res.in/user_manual/National_Polar_Data_Center.pdf	Official NPDC document.
FAIR Principles	Findable, Accessible, Interoperable, Reusable metadata design.	https://www.go-fair.org/wp-content/uploads/2022/01/FAIRPrinciples_overview.pdf	GO FAIR.
Gin	Go REST API framework documentation/repository.	https://github.com/gin-gonic/gin	Official GitHub repository.
pgx	Native Go PostgreSQL driver/toolkit.	https://github.com/jackc/pgx	Official GitHub repository.
PostgreSQL Full Text Search	Database-native text search, ranking and indexing.	https://www.postgresql.org/docs/17/textsearch.html	PostgreSQL official documentation.
PostgreSQL pg_trgm	Fuzzy text similarity and trigram indexes.	https://www.postgresql.org/docs/17/pgtrgm.html	PostgreSQL official documentation.
Groq Quickstart	Current Groq API usage basics.	https://console.groq.com/docs/quickstart	Groq official documentation.
Groq Structured Outputs	JSON-schema constrained model output for supported models.	https://console.groq.com/docs/structured-outputs	Groq official documentation.
Supabase Storage SDK	File storage and access controls.	https://github.com/supabase/supabase-js	Supabase official GitHub repository.
Leaflet Quick Start	Interactive maps, markers and OpenStreetMap attribution guidance.	https://leafletjs.com/examples/quick-start/	Leaflet official documentation.
OpenStreetMap Tile Policy	Operational/attribution requirements for OSM tiles.	https://operations.osmfoundation.org/policies/tiles/	OSM Foundation.
Semantic Scholar API	Paper/author search and scholarly metadata.	https://webflow.semanticscholar.org/product/api	Semantic Scholar official documentation.
Public SIH26063 project: PolaRis	Competitive reference: globe, chatbot, learning, media and roadmap.	https://github.com/Mister-Halder/POLAR-SCIENCE-PORTAL	Public GitHub repository; not an official NCPOR system.
Public PolarNCPOR concept/demo	Competitive reference: auto content generation, smart tagging, scheduling and analytics.	https://polarncpor.vercel.app/	Public web demo; not treated as official NCPOR infrastructure.
Public RAG foundation project	Reference architecture for ingestion, pgvector, hybrid search and source citations.	https://github.com/Ashwin-AIAS/rag-foundation-pgvector	Public GitHub repository; use as architecture reference only.
Go PDF reader	Prototype PDF text extraction	https://github.com/	Public GitHub repository;

Source	Why it matters	URL	Notes
	reference.	ledongthuc/pdf	validate robustness before production use.

30. Final Submission + Demo Checklist

Area	Checklist
Problem	PS ID/title/theme/category match official portal
PPT	Exactly 6 slides using supplied template; PDF export
Solution	Repository + search + expedition + AI + outreach review works
Data	Every demo item has source/rights metadata
AI	Source cards visible; insufficient-evidence behavior tested
Security	No secrets in GitHub/frontend
Frontend	No broken links, no empty pages, responsive enough
Backend	Health endpoint + core CRUD/search/AI routes work
Demo	One coherent story rehearsed end-to-end
Q&A	Team can explain architecture, data flow, AI grounding, scope and limitations
README	Setup + architecture + data sources + limitations included
Git	Final tagged commit created and backed up
Backup	Local demo + deployed demo + screenshots + PPT PDF all available

30.1 “Do not say this” list

- “We have integrated all NCPOR data.” unless you actually have.
- “Our AI cannot hallucinate.”
- “Everything is real-time.”
- “Automatic social publishing is implemented.” unless it is.
- “This is FAIR compliant.” unless formally assessed.
- “We guarantee scientific correctness.”
- “This is the first polar portal.”

30.2 “Say this instead”

- “We built an integration layer for a representative, traceable corpus.”
- “Our generation is source-grounded and review-controlled.”
- “Live connectors are the production roadmap.”
- “The MVP generates platform-ready drafts; publishing remains human-controlled.”
- “We adopt FAIR principles as design guidance.”
- “The original source remains authoritative.”
- “Our differentiator is evidence lineage and workflow, not another isolated archive.”

31. One-Page Team Cheat Sheet

What are we building?

PolarSetu: a unified polar-science discovery and outreach layer.

Why?

To make polar resources easier to find, understand, connect and responsibly disseminate.

Core flow

Search → Resource → Related evidence → Ask AI → Outreach Studio → Human review → Export.

Stack

React/Vite/Tailwind → Go/Gin → PostgreSQL + Storage → Groq; Leaflet for map.

Innovation

Evidence lineage + resource relationships + audience-aware outreach + editorial control.

Do not build

Microservices, Neo4j, Kafka, Elasticsearch, 3D twin, satellite ML, full social automation.

Demo sentence

“We connect polar knowledge into an evidence trail and turn approved science into usable stories.”

End of handbook. This document is intended to be the team's single source of truth for the current 72-hour SIH26063 prototype plan. Re-check current official portal instructions, deadlines, service terms, model availability and source rights before submission.